MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

SAHARA: Real-Time On-Device Sound Alerts for the Deaf and Hard-of-Hearing

A private phone app that hears safety sounds and your name, tuned for few false alarms, with India-specific adaptation

Capstone Project
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Computer Engineering
MPSTME, Mumbai

Abstract, Introduction and Motivation

Over **430 million** people live with disabling hearing loss [1]; about **63 million** in India [2]. A smoke alarm, siren, doorbell, a crying baby or someone calling your name is silent to them unless turned into light or vibration.

SAHARA runs fully on an Android phone: a frozen YAMNet encoder and a small trained head detect **8 household sounds**, a speech model detects **name calls** and who is calling, and every alert is a full-screen notice plus a **class-coded vibration**, on the phone and on a proposed ESP32 wristband. Alerts are tuned to about **4 false alarms per hour**, and we recorded **SAHARA-IN** in Mumbai homes.

430 M	63 M	0
disabling hearing loss (WHO)	in India (NPPCD)	audio files ever stored

Literature Review

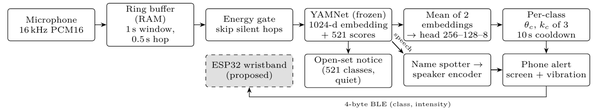
- **Bragg 2016, Findlater 2019:** DHH users want personal, glanceable, wearable alerts.
- **HomeSound 2020:** false alerts quickly erode trust.
- **SoundWatch 2020, ProtoSound 2022:** on-watch recognition; few-shot personal sounds.
- **Huang 2023:** noise and confusable sounds remain open problems.
- **YAMNet / PANNs / AudioSet:** pretrained audio embeddings transfer with little data.

Gap: none tunes alerts to a false-alarm budget, tests in Indian homes, and runs fully offline with name-call alerts.

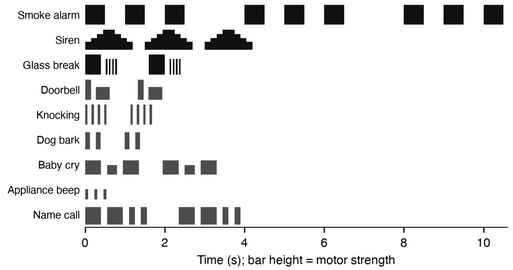
Objectives

1. Detect 8 sounds on the phone in real time.
2. About 4 false alarms/h; missed safety sounds cost 3×.
3. Fully offline: no audio stored, no Internet.
4. Tell users their name is called, and by whom.
5. Build and evaluate an Indian field set.
6. Class-coded vibration on phone and band.

Methods



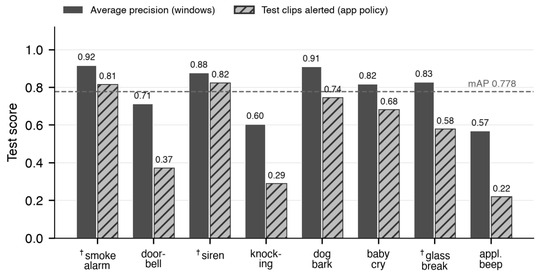
- **On the phone:** 1 s RAM buffer, energy gate, YAMNet embedding every 0.5 s, head 256–128–8 (296k parameters).
- **Alert rule:** per class, score $\geq \theta_c$ in k_c of 3 windows, 10 s cooldown.
- **Budget tuning:** choose every (θ_c, k_c) on validation to maximise safety-weighted detection with ≤ 4 false alarms/h in total.
- **Data:** 22,549 clips from FSD50K, AudioSet, ESC-50, DESED + 329 SAHARA-IN; label IDs verified; splits grouped by recording.
- **Name calls:** speech embeddings + head for “Aryan”; GE2E speaker encoder names enrolled family, else “Someone”.
- **Home calibration:** record a sound 3 times; its threshold adapts, within a tested range.



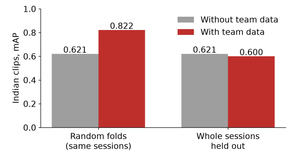
Class-coded vibration patterns, same on phone and band.

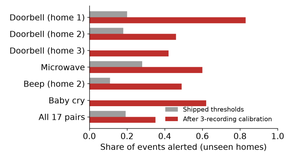
Results

Public test set: 0.778 mAP, macro F1 0.670, **3.5 false alarms/h** on 3.4 h of continuous background.



Alert rule (same model)	Clips	Safety	FA/h
$\theta = 0.5$, 1 window	0.77	0.84	49.1
$\theta = 0.5$, 2 of 3	0.69	0.84	34.1
SAHARA tuned	0.54	0.69	**6.1**





Name call vs. background (AUC)	0.872
Name call vs. similar names (AUC)	0.743
Caller named correctly	29/38
Inference per window (Galaxy S24)	20–29 ms

Discussion

- Budget tuning cut false alarms **34.1** → **6.1 per hour** on the same model.
- Safety sounds are strongest: 22/27 smoke alarms, 28/34 sirens alerted.
- Brief knocks and beeps are weakest.
- Indian recordings help in homes the model has heard (0.62 → 0.82 mAP) but not yet in unseen homes (0.60).
- **Home calibration** lifts unseen homes from 19 % to 35 % of events alerted, about +1 false alarm/h.
- Wristband designed (~Rs. 870), firmware compiles; build and DHH user study are next.

References

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Acknowledgements

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